moves along a straight path with uniform velocity. This path is called mean free path.

Consider a system of molecules each with diameter d. Let n be the number of molecules per unit volume.

Assume that only one molecule is in motion and all others are at rest as shown in the Figure 9.8

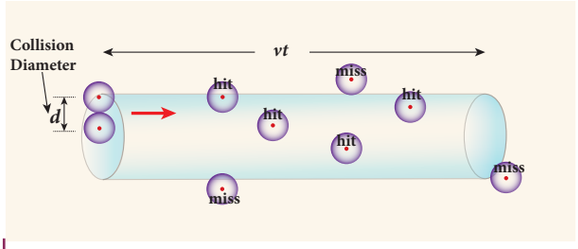


Figure 9.8 Mean free path

If a molecule moves with average speed v in a time t, the distance travelled is vt. In this time t, consider the molecule to move in an imaginary cylinder of volume $\pi d^2 vt$. It collides with any molecule whose center is within this cylinder. Therefore, the number of collisions is equal to the number of molecules in the volume of the imaginary cylinder. It is equal to $\pi d^2 vtn$. The total path length divided by the number of collisions in time t is the mean free path.

Mean free path, $\lambda = \dfrac{distance\ travelled}{Number\ of\ collisions}$

$$\lambda = \frac{vt}{n\pi d^2 vt} = \frac{1}{n\pi d^2} \qquad (9.25)$$

Though we have assumed that only one molecule is moving at a time and other molecules are at rest, in actual practice all the molecules are in random motion. So the average relative speed of one molecule with respect to other molecules has to be taken into account. After some detailed calculations (you will learn in higher classes) the correct expression for mean free path

$$\therefore \lambda = \frac{1}{\sqrt{2}n\pi d^2} \qquad (9.26)$$

The equation (9.26) implies that the mean free path is inversely proportional to number density. When the number density increases the molecular collisions increases so it decreases the distance travelled by the molecule before collisions.

Case1: Rearranging the equation (9.26) using 'm' (mass of the molecule)

$$\therefore \lambda = \frac{m}{\sqrt{2}\pi d^2 mn}$$

But mn=mass per unit volume = ρ (density of the gas)

$$\therefore \lambda = \frac{m}{\sqrt{2}\,\pi d^2 \rho} \qquad (9.27)$$

Also we know that PV = NkT

$$P = \frac{N}{V}kT = nkT$$

$$\therefore n = \frac{P}{kT}$$

Substituting n= $\dfrac{P}{kT}$ in equation (9.26), we get

$$\lambda = \frac{kT}{\sqrt{2}\pi d^2 P} \qquad (9.28)$$

The equation (9.28) implies the following

1. Mean free path increases with increasing temperature. As the temperature increases, the average speed of each molecule will increase. It is the reason why the smell of hot sizzling food reaches several meter away than smell of cold food.
2. Mean free path increases with decreasing pressure of the gas and diameter of the gas molecules.

UNIT-9(XI-Physics_Vol-2).indd 178 26-03-2022 17:47:40